Data Visualization: Assignment 1
Visual EDA and Implementation of Studied Concepts
Deadline: October 19th, 2024 (23:55 PST)

Background:
You have been provided with two datasets; "Airbnb listings dataset" and "Football World cup Matches (1930-2014) records dataset". These datasets offer a rich ground for applying visual exploratory data analysis techniques to uncover patterns, trends, and relationships that could shed light on factors influencing storytelling and Question answering via Visual EDA.

Objectives:
1) Your first task is to conduct a comprehensive visual EDA to explore the potential relationships between attributes present in the dataset. For both of the datasets, your analysis should aim to visually investigate the datasets and create and defend a hypothesis via visualization and its description.
2) Calculate Data-Ink Ratio by using the given visualizations and also comment about each visualization in the light of graphical integrity and visualization principles. Your comments must be thoroughly defended by referring to the specific principles which you think is applied or negated.
3) Create a Graph and network diagram for all football matches of the 2014 world cup via utilization of provided data.

Requirements (Objective - 1):
Data Preparation: Clean and preprocess the data to facilitate effective analysis. This includes handling missing values, correcting data types, and aggregating data where necessary.

Univariate Analysis: Start with visualizing the distribution of individual variables using histograms, density plots, or box plots to understand the variability and presence of outliers.

Bivariate Analysis: Explore the relationships between two variables through scatter plots, correlation matrices, and bar charts. Also perform pair-wise correlation between attributes for both datasets.

Multivariate Analysis: Utilize techniques like Scatter Plot Matrix (SPLOM), Parallel Coordinates, or Trellis Plots to visualize and analyze the relationships among multiple variables simultaneously. You may also Employ other visualization techniques such as glyphs or heatmaps etc. to uncover deeper insights and patterns within the data.

Deliverables:
Visual Analysis Report: A comprehensive report including all visualizations created during the EDA process, accompanied by brief (5 line max) descriptions and interpretations of each plot. Highlight any significant findings, patterns, or anomalies observed in the datasets.

Insights and Conclusions: Conclude your report with a summary of key insights and whether the visual analysis supports or refutes the initial hypothesis. Discuss any unexpected findings and their potential implications.
Recommendations for Further Analysis: Suggest areas for further study or additional variables that could be explored to deepen the understanding of the datasets.

Code files: E.g., "Jupyter notebook" or ". ipynb" files supporting your analysis

Evaluation Criteria:
Completeness: All required sections are thoroughly addressed.

No-Pledge or use of online APIs: No auto-tools are to be used for any part of the assignment.

Analytical Depth: Demonstrates a deep understanding of visual EDA techniques and their application to uncover insights.

Insightfulness: Ability to derive meaningful conclusions from visual analyses.

Presentation: Clarity and effectiveness of visualizations and accompanying explanations.

Note: This assignment is to be done *individually*. Plagiarism of any kind ***will*** result in ***0*** marks straightaway which will result in deduction of '***5' Solid*** Marks. You are not to use any APIs or auto-EDA tools for this assignment.